

Method Review and Costing

2026.08.18 - Method Review and Costing for Microbial Perturb-seq

Companion design sketch: `[[scratch.2026.08.14.132048-spc-perturb-seq]]` (Atrandi SPC + CapSeq route). This note covers the **commodity, benchtop route** – split-pool and droplet – and prices it end to end.

All tables also render standalone (landscape, one per block, for screenshotting): `[[microbe-perturb-seq.tables]]` -> `notes/assets/pdf-output/microbe-perturb-seq.tables.pdf`

Generating scripts (every number below is regenerated by these; nothing is hand-authored):

- `experiments/019-perturb-seq-costing/scripts/method_data.py` – literature constants
- `experiments/019-perturb-seq-costing/scripts/cost_data.py` – Brettner + Gaisser line items
- `experiments/019-perturb-seq-costing/scripts/uiuc_core_data.py` – UIUC Carver rates
- `experiments/019-perturb-seq-costing/scripts/read_structure.py` – read configurations
- `experiments/019-perturb-seq-costing/scripts/cost_model.py` – the scenario engine
- `experiments/019-perturb-seq-costing/scripts/cost_per_cell_table.py` – cost-per-cell tables
- `experiments/019-perturb-seq-costing/scripts/plot_method_landscape.py, plot_economics.py`

Sources – sha256-verified artifacts from the `torchcell-library` mirror via `tc-lit`:

- `brettnerUltraHighthroughputMassively2024` – yeast SPLiT-seq, the cost breakdown
- `jarianiNewProtocolSinglecell2020` – 10x + in-droplet zymolyase, mRNA per cell
- `nadal-ribellesRiseSinglecellTranscriptomics2024` – yeast scRNA-seq review
- `urbonaiteYeastoptimizedSinglecellTranscriptomics2021` – table-top yeast Drop-Seq
- `brandnerPooledSinglecellCRISPRa2025` – mapSPLiT, pooled CRISPRi/a in bacteria
- `gaisserHighthroughputSinglecellTranscriptomics2024` – microSPLiT protocol + costs
- `sunGenomescaleCRISPRiScreening2023` – CRISPRi library review
- `lianMultifunctionalGenomewideCRISPR2019` – MAGIC, the library we hold

Gaisser’s cost supplement is not in the mirror and was pulled separately from Springer ESM (41596_2024_1007_MOESM2_ESM.xlsx sha256 2b7813a7...).

Bottom line

1. **The \$10,000 “start-up cost” in Brettner is one purchase order: the custom IDT barcode plates.** \$7,699.40 of it, and Gaisser’s independent microSPLiT accounting puts the same item at \$7,844.52 – two labs, two organisms, 1.9% apart. It is a **capital asset good for 215 protocol runs**, not a consumable. You will never exhaust it; oligo degradation will retire it first.
2. **The advertised cheapness of SPLiT-seq is a reagents-only number and it does not survive contact with a sequencer.** Brettner’s \$0.005/cell becomes **\$0.104 per usable cell** end to end – a 21x hidden multiplier – because 93.7% of recovered transcripts are rRNA and ~75% of sequenced cells get discarded. The honest split-pool-vs-droplet gap is **2.3x, not 60x**.
3. **rRNA depletion is the single highest-leverage change available.** It cuts the genome-scale bill from **\$157k to \$58k** and is the difference between “we should think about this” and “we can afford this”.
4. **A genome-scale screen needs a 4th barcoding round.** Not for throughput – for collisions. At 12M barcoded cells, three rounds plus 96 sublibraries still collide at 13.2%; four rounds plus 24 sublibraries collide at 0.59%. The 4th plate costs ~\$2,600 once and **pays for itself inside one run**.
5. **The blocking problem is not cost, it is that the guide is invisible.** The MAGIC library we already have is RNA Pol III (SNR52p-guide-SUP4t), so it is not polyadenylated and **cannot be read by any 3’ capture assay**. Every platform here needs an expressed poly(A) barcode added. This is the real critical path.

1. How much mRNA is in a yeast cell

The denominator for everything else.

Organism	Quantity	Value	Source
<i>S. cerevisiae</i>	Total mRNA molecules per cell	20,000 – 60,000	Jariani 2020 (paper.md:31), citing von der Haar 2008, Marinov 2014, Pelechano 2010, Zenklusen 2008
<i>S. cerevisiae</i>	Working figure used by Brettner	~30,000	Brettner 2024 (paper.md:70), citing Miura 2008
<i>S. cerevisiae</i>	Total RNA per cell	0.7 – 1 pg, ~85% rRNA	Nadal-Ribelles 2024 (paper.md:49)
<i>S. cerevisiae</i>	Genes expressed in a basal cell	~3,000 (half the genome), avg 3.5 molecules/gene	Nadal-Ribelles 2024 (paper.md:107)
<i>E. coli</i>	Total mRNA per cell, rich medium (LB)	~7,800	Bartholomäus 2016, BNID 112795 (external)
<i>E. coli</i>	Total mRNA per cell, minimal medium	~2,400	Bartholomäus 2016, BNID 112795 (external)
<i>E. coli</i>	Total mRNA per cell, medium growth rate	~3,000	Taniguchi 2010, <i>Science</i> 329:533 (external)
<i>E. coli</i>	rRNA share of all transcripts	>95%	Brandner 2025 (paper.md:71)
Mammalian	Total mRNA molecules per cell	50,000 – 300,000	Jariani 2020 (paper.md:31)

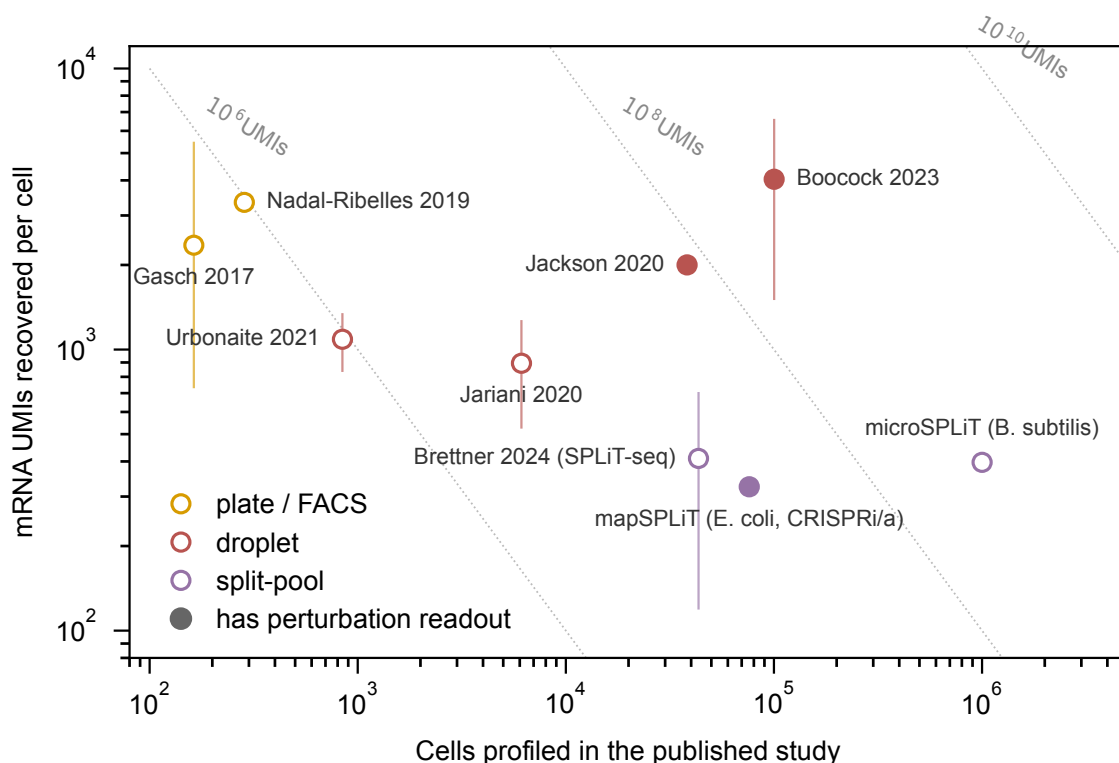
The *E. coli* rows are marked (*external*) because they are NOT in the mirror. No mirrored bacterial paper states a total mRNA content – Gaisser and Brandner report only what their assay *detected*, which is a different quantity and ~30x smaller. Pull the primary paper into `torchcell-library` before these enter a proposal.

Flagged disagreement. Nadal-Ribelles’ 3,000 genes x 3.5 molecules implies ~10,500 mRNA/cell, materially below Jariani’s 20,000 floor. Both are cited literature. The model uses 30,000 and the conclusions are not sensitive to the choice, but do not quote a single value as settled.

The consequence that matters: a yeast cell has **~5x less mRNA than a mammalian cell spread over ~3x fewer genes**. Every scRNA-seq chemistry in existence was tuned on mammalian input. That is why yeast per-cell recovery is poor everywhere, and why pooling cells is not a convenience but the whole measurement strategy.

And a yeast cell holds ~4x the mRNA of an *E. coli* cell in LB, ~10x one in minimal medium. That, plus the worse bacterial rRNA burden (>95% vs ~85%), is why the same split-pool chemistry recovers 410 mRNA UMIs/cell in yeast but only 235–397 in bacteria. Yeast is the easier organism here, which is worth remembering given that the only published pooled single-cell CRISPR screen is bacterial.

2. The method landscape



Dotted diagonals are iso-lines of constant **total** mRNA UMIs. In the shot-noise limit that product – cells x depth – is what sets precision, so moving along a diagonal is free and moving up-and-right is what costs money. Filled markers have a perturbation readout.

Study	Platform	Isolation	Cells	mRNA UMIs/cell	Genes/cell
Gasch 2017	Fluidigm C1 / SMART-Seq v4	plate	163	2,351 median (735–5,437)	–
Nadal-Ribelles 2019	FACS + STRT-seq	plate	285	3,339 mean	–
Jackson 2020	10x 3' v2	droplet	38,225	~2,000	~684–700
Jariani 2020	10x 3' v2 + in-droplet zymolyase	droplet	6,118	528–1,261 median	–
Urbonaite 2021	yeastDrop-Seq (table-top)	droplet	844	840–1,336 mean	443–619
Dohn 2021	mDrop-seq	droplet	74,814	–	193–967
Boocock 2023	10x 3' v2 + v3	droplet	100,220	1,514–6,559	–
Brettner 2024	yeast SPLiT-seq	split-pool	43,388	120–700 mean	100–600
microSPLiT (<i>B. subtilis</i>)	microSPLiT	split-pool	~1,000,000	397 median	230
mapSPLiT (<i>E. coli</i> , CRISPRi/a)	microSPLiT + guide barcode	split-pool	76,068 assigned	325 median	161

Two corrections to the naive reading:

- **Nadal-Ribelles 2019 is reported inconsistently.** Jariani says “detected on average 3339 transcripts”; the Nadal-Ribelles review’s own Table 1 says “3399 genes”. 3,399 genes in one yeast cell is not plausible for scRNA-seq. Treated here as transcripts.
- **Brettner and Boocock are the same era but an order of magnitude apart in depth.** Boocock got 1,514–6,559 UMIs/cell on 100,220 cells with stock 10x v3. That is the real frontier for yeast, and it is droplet, not split-pool.

3. Anatomy of the Brettner cost model

All quotes from `brettnerUltraHighthroughputMassively2024 Methods 5.1.9 (paper.md:183)`.

3.1 What the start-up cost actually is

“300uL 100uM prediluted plates for the round 1 reverse transcription and rounds 2 and 3 ligations were ordered from Integrated DNA Technologies (IDT) for **\$7699.40**. These plates contain enough volume to complete **300 RT, 250 r2, and 215 r3 ligations** with careful pipetting. Using the lowest capacity of the round 3 plate, this comes to **\$36 in barcoding oligos per protocol**.”

That is the answer to “what is the upstart cost referring to”. It is **three 96-well plates of custom barcoded oligos**, and it is ~77% of the \$10,000. The rest:

Item	Cost	Scaling
IDT barcode plates (RT r1, ligation r2, r3)	\$7,699.40	one-time, 215 runs
Standalone oligos (lyophilized)	\$270.60	one-time
Maxima H Minus RT (Thermo EP0753)	\$835–845	per run (“enough for just over 1 experiment, but not enough for 2”)
T4 DNA Ligase (NEB M0202M)	\$270	per run
RNase inhibitors (SUPERase-In + Enzymatics)	\$630 / 5 runs = \$127	per run
Barcoding oligos drawn from the plates	\$36	per run
Dynabeads MyOne Streptavidin C1	\$638/2 mL, 44 uL each = \$14	per sublibrary
Enzymatics WGS fragmentation + ligation	\$39	per sublibrary

Rolled up by Brettner: **start-up ~\$10,000, per-protocol ~\$1,300–1,700, plus \$55 per sublibrary**. (The OCR renders the middle figure as “\$12300-1700”; summing the line items gives $840 + 270 + 127 + 36 = \$1,273$, confirming \$1,300–1,700 is intended.)

Independent replication. Gaisser’s microSPLiT supplement, built separately for bacteria, gives start-up **\$8,142.52**, of which **\$7,844.52** is “Barcode Plates, 100uM, Custom IDT” (sic). Per run: \$790.78 for 1 sample, \$2,035.82 for 24, \$3,196.53 for 48 – and critically, **barcode cost per run is flat at \$20.67 regardless of sample count**. Two independent accountings agree that the barcode plates are the start-up cost and that their per-run draw-down is trivial.

3.2 What constitutes a “full experiment” One **protocol run** (Brettner’s “instance of the protocol”) is one pass through:

1. fix + permeabilize cells, count, dilute to 1M cells/mL;
2. load **up to 96 samples** into the RT plate (one barcode per well);
3. two ligation rounds with pooling and re-splitting between;
4. pool, split into **sublibraries**, lyse;
5. template switch, cDNA amplification, WGS library prep **per sublibrary**;
6. sequence.

It consumes exactly one use of each barcode plate, one tube of RT enzyme, one tube of ligase, and one fifth of the RNase inhibitors. Brettner ran RT in **48 wells at 5,000 cells/well = 240,000 cells**; scaled to a full 96-well plate that is **~480,000 cells per run**, which is where the paper’s “approximately 400,000 cells for approximately \$2000” comes from.

The **sublibrary** is the separate unit: it is the unit of library prep, of Illumina index, and (see §4) of barcode space. Brettner split into 10 sublibraries of 5,000–20,000 cells.

Cell recovery is poor and must be budgeted: “Given we started the experiment with between 200,000 and 300,000 cells, this estimates a capturing efficiency between **25% and 50%**” (paper.md:68). Gaisser is blunter – cell loss “could reach $\geq 70\%$ ”, and they refuse to run below 1M input cells.

3.3 When does the start-up cost get repurchased Effectively never, on volume. The round-3 plate is the binding constraint at 215 uses. At a realistic 5–10 protocol runs per year that is a **20-year supply**. The practical retirement trigger is **oligo degradation and freeze-thaw**, not depletion.

Two consequences worth acting on:

- **You are massively over-buying.** 300 uL at 100 uM per well is ~30 nmol per oligo. If the plan is 20 runs, not 215, ask IDT for a lower synthesis scale or smaller volume. *This has not been priced – get a quote before assuming a large saving, since plate and per-oligo synthesis fees may dominate volume.*
- **It is a natural thing to share.** One plate set supports an entire department.

The genuinely recurring items are the RT enzyme (\$840) and ligase (\$270), which Brettner explicitly repurchases every run.

3.4 Cost per cell, with citations Generated by `cost_per_cell_table.py -> results/cost_per_cell.csv`.

Method	Organism	Isolation	\$ / cell	Basis	Source
MATQ-seq	<i>bacteria</i>	plate	\$50.00	derived	Gaisser 2024, <i>Nat Protoc</i> , Suppl. Table 2
Plate FACS + STRT-seq (Nadal-Ribelles 2019)	<i>S. cerevisiae</i>	plate	\$4.15	quoted	Jariani 2020, <i>eLife</i> 9:e55320
10x Chromium 3' v2 (Jariani 2020)	<i>S. cerevisiae</i>	droplet	\$0.3000	quoted	Jariani 2020, <i>eLife</i> 9:e55320
ProBac-seq	<i>bacteria</i>	droplet	\$0.1649	derived	Gaisser 2024, <i>Nat Protoc</i> , Suppl. Table 2
PETRI-seq	<i>bacteria</i>	split-pool	\$0.0367	derived	Gaisser 2024, <i>Nat Protoc</i> , Suppl. Table 2
microSPLiT (1 sublibrary)	<i>bacteria</i>	split-pool	\$0.0172	derived	Gaisser 2024, <i>Nat Protoc</i> , Suppl. Table 2
SPLiT-seq (Brettner 2024)	<i>S. cerevisiae</i>	split-pool	\$0.00500	derived	Brettner 2024, <i>Yeast</i> 41:242
microSPLiT (full scale)	<i>bacteria</i>	split-pool	\$0.00251	derived	Gaisser 2024, <i>Nat Protoc</i> , Suppl. Table 2

Verbatim anchors for the two **quoted** rows (the rest are a total divided by a cell count the same source supplies, and are marked derived):

“the ability to increase sample size in droplet-based methods allows the library preparation cost to be greatly reduced to **\$0.3/cell** compared to previously reported costs of **\$4.15/cell** (Nadal-Ribelles et al., 2019)” – Jariani 2020, `paper.md:172`

The spread is ~20,000x. It is also misleading, because **every row is library prep only**. Costing the same platforms end to end, per cell that survives QC *and* carries an identified guide (`results/cost_per_cell_loaded.csv`, 6,000 genes at 250 cells/gene):

Platform	Reagents \$/cell processed	Loaded \$/usable cell	Hidden multiplier	Sequencing share
SPLiT-seq, as published	\$0.0049	\$0.1044	21.3x	81.2%
SPLiT-seq + rRNA depletion	\$0.0053	\$0.0383	7.2x	44.2%
10x Chromium X (GEM-X 3')	\$0.1077	\$0.2424	2.3x	19.2%

The model reproduces Brettner’s \$0.005/cell from independent line items (\$0.0049), which is the check that it is wired up correctly.

This table is the most important result in the note. The reagents-only comparison says split-pool beats droplet by 22x. The real comparison says **2.3x**. The gap is rRNA and discarded cells, and it is recoverable – with depletion the advantage returns to 6.3x.

4. Barcode space: three rounds, four rounds, and sublibraries

Collision rate, per Gaisser’s published formula $100(1 - ((B - 1)/B)^{C-1})$ with B barcode trajectories and C cells barcoded together (`results/barcode_collision.csv`; the model reproduces their 5%-at-45,000-cells operating point to 4.96%).

The commonly missed point: the sublibrary index is a real barcode dimension. Cells are split into sublibraries *after* the last in-cell ligation, so two cells sharing a BC1-BC2-BC3 path are still resolved if they land in different sublibraries. Effective space is $96^R \times S$, not 96^R . Leaving S out is the usual way to over-estimate collisions – and including it is how Brettner’s ~880k barcode space supported 43,388 cells at “well under five percent” collisions.

Rounds	Sublibraries	Barcode space	Collision @ 480k cells	Collision @ 12M cells
3	1	884,736	41.9%	100%
3	24	2.12e7	2.2%	43.2%
3	96	8.49e7	0.56%	13.2%
4	24	2.04e9	0.02%	0.59%
4	96	8.15e9	0.006%	0.15%

How “4x barcodes” changes the pricing structure Two readings, both real, and they price very differently.

(a) **A fourth barcoding round** – what Brettner means by “adding one more ligation reaction yields almost 85 million unique combinations with only the trivial cost of adding another plate of 96 barcode oligos” (`paper.md:46`), and what microSPLiT already does as standard.

- **One-time:** one more IDT plate, ~\$2,566 (pro-rata from the \$7,699.40 three-plate order).
- **Per run:** ~\$12 more barcode draw-down, plus ~50% more T4 ligase (~\$135) for the third ligation. Call it **+\$150/run**.
- **Buys:** 96x the barcode space.
- **Costs:** one more split-pool ligation, and every ligation loses material. Yield loss is the only argument against and must be measured in the pilot.

The payback is under one run. At 480,000 cells/run and a 1% collision target, three rounds needs **54 sublibraries** (\$2,970/run in Dynabeads + WGS kits). Four rounds clears 1% at a single sublibrary, so the count drops

to whatever PCR complexity dictates – ~ 11 at Gaisser’s 45,000-cells-per-sublibrary working limit (\$605/run). That saves $\sim \$2,365$ per run against a \$2,566 one-time cost.

(b) **384-well plates instead of 96** – $384^3 = 5.7 \times 10^7$, comparable to 96^4 , and it quadruples RT-plate throughput to $\sim 1.9\text{M}$ cells/run. But it costs **4x the oligos** ($\sim \$30,000$ for the plate set) and 4x the pipetting. For a pooled screen where the RT barcode is *not* being used for sample multiplexing, this is the way to cut the number of protocol runs – but the 4th ligation round is far cheaper per unit of barcode space.

Structural point for a pooled screen. Brettner sells SPLiT-seq on “96 samples multiplexed in one run”. In a pooled CRISPRi screen you have **one** sample; guide identity comes from the transcriptome, not the well. So the RT plate’s value flips from multiplexing to raw throughput, and the multiplexing capacity should be spent on **environments** instead – which is the axis that actually multiplies design size.

5. How many genes per cell does the method allow

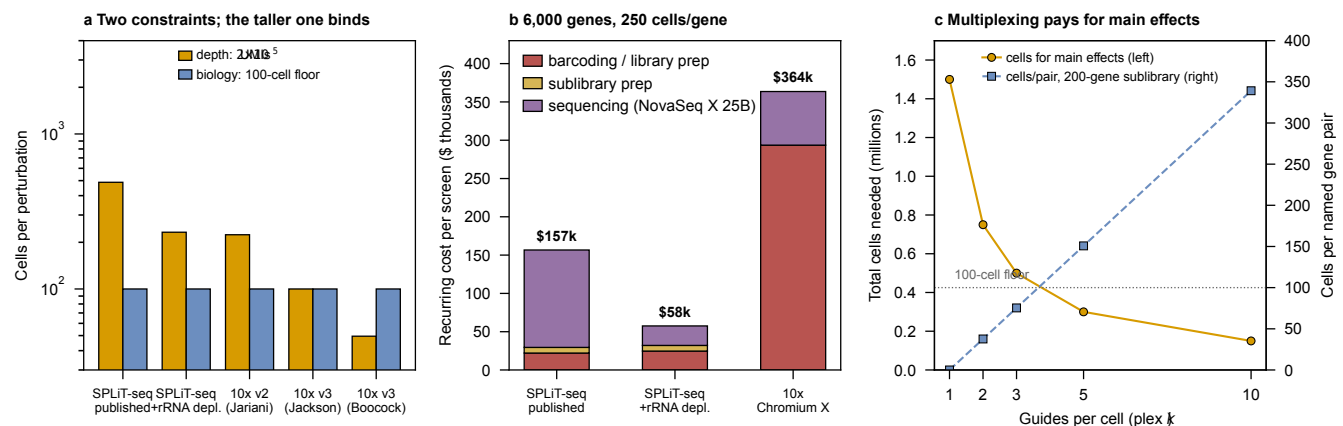
The question has two readings and both matter.

(a) **Genes detected per cell.** Brettner: **100–600 on average, 120–700 mRNAs**, i.e. “0.2-12% of the active transcriptome”. Against $\sim 6,000$ genes, **$\sim 90\text{--}98\%$ of the single-cell matrix is zero.**

This is the uncomfortable consequence: at these depths you cannot read one cell’s transcriptome. The as-say is really a **massively multiplexed pseudobulk** measurement with the *option* of stratifying by cell state, not a single-cell assay in the sense the name implies. Brettner says as much – “algorithms leverage the fact that the transcripts recovered differ for every cell, so they can construct a fuller transcriptome by clustering together many single-cell transcriptomes”. Plan the analysis around pseudobulk-per-guide from the start.

(b) **Genes perturbed per cell.** As published, **one**. Multi-guide is demonstrated in the same chemistry by Brandner: “Two or three genes were targeted simultaneously by expressing multiple sgRNAs from a plasmid containing a single GBC ... ten two-guide cassettes and eight three-guide cassettes”. The ceiling is set by cloning and by guide detection compounding (§9), not by the barcoding chemistry.

6. How many cells per guide to see the mean behaviour



Two **independent** constraints. The taller one binds (panel a).

Constraint 1 – biological floor. Averaging out cell-cycle and metabolic-state heterogeneity takes cells, and no amount of sequencing depth substitutes. The field standard, stated explicitly by Brandner: perturbations are kept only with “**at least 100 cells** following standard approaches in scRNA-seq perturbation screens” (paper.md:89). Brettner’s own heuristic is “500 cells per experiment to achieve adequate coverage” (paper.md:48).

Constraint 2 – pseudobulk depth. For gene j pooled over n cells at depth d , pseudobulk counts are $\approx ndp_j$ and $\text{Var}(\log K) \approx 1/K$. Detecting a fold change Δ at 80% power, $\alpha = 0.05$ needs

$$K \geq \left(\frac{z}{\Delta \ln 2} \right)^2, \quad z = 2.80,$$

so ~**16 UMIs of that gene** for 2-fold, ~65 for 1.4-fold, ~159 for 1.25-fold. What a given pseudobulk budget reaches (`results/detection_ladder.csv`):

Tier	Pseudobulk mRNA UMIs / perturbation	Lowest detectable gene, 2-fold	... at 1.5-fold
Screen (strong coherent effects)	5.0e4	9.8 copies/cell	28.6 copies/cell
Standard	2.0e5	2.5 copies/cell	7.2 copies/cell
Deep (low-abundance regulators)	1.0e6	0.5 copies/cell	1.4 copies/cell

At the standard tier, cells needed per perturbation = $2 \times 10^5/d$:

Method	mRNA UMIs/cell	Cells/perturbation	Binding constraint
SPLiT-seq as published	410	488	sequencing depth
SPLiT-seq + rRNA depletion	861	233	sequencing depth
10x v2 (Jariani)	894	224	sequencing depth
10x v3 (Jackson)	2,000	100	biological floor
10x v3 (Boockock)	4,036	100	biological floor

Two things fall out. First, **488 cells reproduces Brettner's own 500-cell heuristic** from an independent derivation, which is a good sign the framework is right. Second, **depth buys you out of cells only until you hit 100**; past 10x v3 the extra depth is wasted on this question and should be spent on more perturbations instead.

7. Sequencing: what UIUC actually has, and single vs paired end

Rates from <https://biotech.illinois.edu/dna-services-core/dna-services-pricing/>, effective **2026-08-01**, retrieved 2026-08-18. Listed rates are the on-campus rate; external users pay +30.8% (capped at 20% for State of Illinois institutions).

The instrument list is narrower than you may expect. The core has **only the NovaSeq X Plus and the MiSeq i100** (plus PacBio Revio and Nanopore GridION). There is **no NextSeq and no NovaSeq 6000** – two pages still mention a 6000 in stale prose, but it is not on the instrument list or the rate table. This matters because the microSPLiT protocol is written against a NextSeq 2000 P2, which does not exist here. The real choice is a ~25M-read MiSeq run or a >=800M-read NovaSeq X lane, with **nothing in between**.

Single end or paired end Both methods require paired-end. Neither can use a single-read flow cell. But they put the barcode on *opposite* reads, which trips people up at submission (`read_structure.py`):

Method	Read 1	Read 2	Index	Cycles
SPLiT-seq	74+ nt cDNA	86 nt barcodes	6 or 8 nt	160
10x 3'	28 nt barcode	90 nt cDNA	dual 10 nt	118
Drop-seq	20 nt barcode	100 nt cDNA	8 nt	120

Read 2 in SPLiT-seq is BC3 + BC2 + BC1 + a 10 nt UMI; Read 1 in 10x is a 16 nt cell barcode + 12 nt UMI (v2 is 16 + 10). **The SPLiT-seq index is the sublibrary index, which is barcode round 4** – not merely a sample tag.

Gaisser is explicit: “Use a paired-end sequencing run with a >=150-bp kit. Set read2 to 86 nt (cell-specific barcodes and UMI). Set read1 to >=74 nt (cDNA sequence)... Include a 6-nt read 1 index to read sub-library indices (this is the fourth round of barcodes). Use [>=]10% PhiX spike-in.” Brettner: “Experiment 1 was paired end sequenced on a full lane of the HiSeq X ... Experiments 2 and 3 ... with dual 8 bp indices.”

Neither can run single-read for the same reason: with one read you get barcodes and no transcript, or transcript and no cell identity. Note the asymmetry – for 10x the cDNA itself is still sequenced from one end only; “paired” there means “barcode read plus insert read”.

Paired-end costs nothing extra at UIUC, which is not obvious and is worth checking at any other core:

- Nova X 25B PE150, 8+ lanes: **\$3,180 / 3.2e9 pairs = \$0.994 per M read pairs**
- Nova X 10B SR100, 8+ lanes: \$1,270 / 1.2e9 reads = \$1.058 per M reads

The largest flow cell is paired-end only and still wins on price per read.

Three sequencing gotchas specific to split-pool

1. **PhiX is a 10% tax, not hygiene.** Read 2 carries fixed linker sequence at defined cycles, so base diversity collapses there and patterned flow cells mis-register clusters. Budget $\geq 10\%$ of every lane as PhiX. Combined with 70–90% valid-barcode rates, only **~72% of purchased reads become usable data** – this is in the model.
2. **Unique dual indices are mandatory on NovaSeq X.** ExAmp clustering on a patterned flow cell hops indices at ~0.1–2%. For ordinary RNA-seq that is minor sample bleed. For SPLiT-seq the sublibrary index **is barcode round 4**, so a hopped index moves a cell into another sublibrary’s namespace and manufactures a barcode collision. Brettner moved from a single 6 bp index to dual 8 bp indices between experiments.
3. **Spend the free cycles on cDNA.** A 2x150 kit gives 300 cycles; you need 160. Set read 1 to 150 nt rather than 74. It is free and it helps with the *S. cerevisiae* whole-genome-duplication paralogs that forced Brettner to keep multimappers.

MiSeq vs NovaSeq: quality is not the difference Base quality is comparable (MiSeq is if anything slightly better on long reads). The difference is **~1,000x in read count**, so the two have different jobs:

Use	Instrument	Cost
Guide-library representation check (pre-transformation)	i100 25M SR100	\$986
Barcode-structure / complexity QC on one sublibrary before committing a flow cell	i100 titration run	\$398
Saturation curve to fix per-cell read depth	i100 25M 2x150	\$1,450
Production	Nova X 25B PE150, 8+ lanes	\$3,180/lane

Always burn one \$398 titration run before a NovaSeq lane. Brettner’s warning is that there is no earlier checkpoint: “cDNA gel traces or library concentrations are not a good indicator of barcoding success or of the number of reads or genes that will be recovered per cell”. A \$398 run protects a \$3,180–\$25,440 commitment.

The other NovaSeq consideration is **granularity**. A 25B lane is 3.2 billion read pairs. If the experiment does not fill it, you either co-load with other users or waste the difference – the 10B **split lane** (\$1,160) is the smallest sensible production unit.

Cell counting and concentration QC

- **Cell counter: Nexcelom (Revvity) Cellometer K2**, brightfield + AO/PI live/dead. And it is free: “There is no charge to run test counts, either independently or with assistance from DNA Services employees. **None.** All consumables for counting are also included.” Viability gate is $>70\%$, ideally $>90\%$. CMtO separately has a **Bio-Rad TC20** at \$3.30/slide (the same counter Jariani used).
- **Concentration: Qubit fluorometry (\$7) plus Fragment Analyzer** (\$118 for 1–12, \$449 for 96) and qPCR. **They do not have a Qiagen QIAxpert, a NanoDrop, a TapeStation, or a Bioanalyzer service.** If the Zhao lab has a QIAxpert, that is a lab instrument, not a core one – and for library QC, Qubit + Fragment Analyzer is the right pair anyway, since absorbance-based quantification cannot distinguish library from adapter-dimer.
- Cell counting accuracy is not a side issue for split-pool: cells/well sets the collision rate directly, so a systematic miscount silently degrades the whole run.

10x at UIUC Chromium X, GEM-X chemistry, and **CRISPR screening is an offered add-on**: “Add-ons to the RNASeq libraries include V(D)J, Cell Surface Protein, and CRISPR Screening”, at **\$263/sample** on top of the library. Library prices: \$2,650 for one, **\$1,880 each at 8+ submitted together**. Core guidance is “ $\geq 50k$ sequencing reads per targeted cell” (mammalian-calibrated; a yeast cell has $\sim 1/5$ the mRNA, so the model uses 20,000 and flags the deviation) and “ $\sim 8\%$ [doublets] at 20,000 cells with the latest GEM-X kits”.

Turnaround: 10x library construction is “ ~ 4 -8 weeks from when you contact us”, general NGS “one to three weeks”.

8. What a 6,000-gene CRISPRi perturb-seq actually costs

Design: 6,000 genes, MAGIC-style 6 guides/gene (36,100 elements incl. controls), one environment. Full grid in `results/genome_scale_budgets.csv`; panel b of the figure.

Platforms: **SP** = SPLiT-seq as published, **SP+d** = SPLiT-seq with rRNA depletion, **10x** = Chromium X. “Runs” is protocol runs for SP, channels for 10x.

Cells/gene	Plat.	Usable	Seq'd	Runs	Read pairs	Reagents	Seq.	Total
100	SP	600k	2.4M	5	50.0e9	\$11,470	\$50,880	\$62,350
100	SP+d	600k	2.4M	5	10.0e9	\$12,470	\$12,720	\$25,190
100	10x	600k	1.09M	55	27.3e9	\$117,865	\$28,620	\$146,485
250	SP	1.5M	6.0M	13	125e9	\$29,470	\$127,200	\$156,670
250	SP+d	1.5M	6.0M	13	25.0e9	\$32,070	\$25,440	\$57,510
250	10x	1.5M	2.73M	137	68.2e9	\$293,591	\$69,960	\$363,551
500	SP	3.0M	12.0M	25	250e9	\$57,185	\$251,220	\$308,405
500	SP+d	3.0M	12.0M	25	50.0e9	\$62,185	\$50,880	\$113,065
500	10x	3.0M	5.45M	273	136e9	\$585,039	\$136,740	\$721,779

Add **~\$10,000 one-time** platform start-up for the split-pool rows, plus library construction (\$10). Not included: labour, oligo pool synthesis, strain construction.

Reading the table.

- **The dominant term moves.** For split-pool as published, sequencing is **81%** of spend. For 10x it is 19% and library prep is 81% – 137 channels at \$2,143 each. These are different businesses and the optimisations do not transfer.
- **rRNA depletion is worth ~\$99,000** at the 250-cells/gene tier and roughly a third of that even at the 100-cell tier. Nothing else on this page is close.
- **13 protocol runs is the hidden cost of the split-pool row.** That is ~ 13 weeks of skilled bench work, each run a chance to fail, against 137 channels the core runs for you. The \$306k gap between SPLiT-seq-depleted and 10x buys a lot of technician time. This is a real argument for 10x that the dollar column hides.
- **You pay to sequence cells you throw away.** 6.0M sequenced for 1.5M usable. Brandner measured this directly: 365,000 sequenced, 76,068 retained.

Recommended entry point: 100 cells/gene, SPLiT-seq with rRNA depletion, ~\$25k recurring + ~\$10k start-up. That is a genuine genome-scale yeast perturb-seq for under \$40k of consumables and sequencing, and it is defensible because 100 cells/gene is the published field standard.

9. Multiplexing: the end goal

Panel c. For metabolic-engineering strains carrying several perturbations, multiplexed guides are the target, and the arithmetic is unusually clean.

Main effects get k times cheaper. With random k -subsets of T genes and N cells total, the cells informing gene g 's additive main effect are Nk/T – **independent of how many distinct combinations the library holds**. So:

Plex k	Cells for 250-cells/gene main-effect precision	Cells per <i>named</i> gene pair (6,000 genes)	... in a focused 200-gene sublibrary
1	1,500,000	0	0

Plex k	Cells for 250-cells/gene main-effect precision	Cells per <i>named</i> gene pair (6,000 genes)	... in a focused 200-gene sublibrary
2	750,000	0.042	37.7
3	500,000	0.083	75.4
5	300,000	0.167	150.8
10	150,000	0.375	339.2

Named interactions do not become reachable genome-wide at any k . There are 1.8×10^7 gene pairs; random 3-plexing over 6,000 genes puts **0.083 cells** on a specific pair. Sun’s review makes the same point about library construction alone: “constructing a dual-spacer gRNA library for the *E. coli* genome ... would require at least 1.6×10^7 elements”. Lian hit it empirically with sMAGIC – only “~0.01% to ~0.1% of the total combinations” were covered, and they attribute the failure to find better combinations to exactly that.

So the design splits in two:

- **Genome-wide, $k = 3$:** buys main effects at a third of the cell cost. 500,000 cells instead of 1.5M. This is the cheap, correct way to run the genome-scale arm.
- **Focused sublibrary, 100–200 metabolic genes, $k = 2$ – 3 :** 19,900 pairs at ~75 cells/pair is reachable inside one run. This is where named epistasis actually comes from, and it is where torchcell’s inference has training signal for the combinations you never measure.

Three costs of multiplexing that the cell arithmetic hides:

1. **Guide detection compounds.** Assigning k guides at per-guide detection q succeeds at q^k . Brandner’s *E. coli* assignment rate was **25%** even with 5x enrichment primer (“Without enrichment, we were able to correctly assign only 5.5% of cells”). At $q = 0.6$, a 3-plex assigns at 22%.
2. **Collisions become false interactions.** A barcode collision or multiplet in a singles screen attenuates a main effect by a few percent. In a multiplex screen it **manufactures a gene pair that was never made**. Interactions are exactly what you are trying to measure, so the tolerable collision rate is far lower – another argument for the 4th barcoding round.
3. **Interaction terms need ~3x the cells of a main effect** for equal precision ($\varepsilon = \theta_{AB} - \theta_A - \theta_B$ sums three variances), and interactions are typically smaller than main effects, so ~12x is realistic.

10. The blocking problem: the guide is invisible

This is the real critical path, and it is not a cost problem.

MAGIC guides cannot be read by any 3’ capture assay. The library (lianMultifunctionalGenomewideCRISPR2019) expresses guides from SNR52p – guide – SUP4t: **RNA Pol III**, terminating on a T-tract, **not polyadenylated and not capped**. Lian’s own scoring function penalises polyT inside the spacer precisely because “PolyT may be read as a terminator by the Type III RNA polymerase”. Their screen readout is plasmid DNA amplicon sequencing, not RNA. For CRISPRd the variable region is 121 nt (100 nt donor + 21 nt guide), too long for a short-read 3’ tag even if it were polyadenylated.

Nor do the mammalian workarounds transfer. 10x direct capture needs a modified sgRNA scaffold carrying a capture sequence; CROP-seq relies on the guide cassette being duplicated into a lentiviral 3’ LTR, a retroviral trick unavailable in yeast.

The fix is an expressed, polyadenylated barcode, and it is well-precedented:

- Brandner’s **GBC**: “a 316 bp transcript with a unique 12 bp barcode encoding the perturbation identity at the 5’ end”, with a poly(A) tail and a dedicated enrichment primer site, using a dCas12a gene body as inert filler. Enrichment at 5x primer concentration lifted correct assignment from **5.5% to 21–25%**.
- Jackson 2020 did the yeast version: barcoded deletion strains with “a genotype-specific barcode at the 3’-UTR of the marker, captured by 3’ end libraries”.
- The review states the gap plainly: “in yeast, there is a technological gap to perform these analyses because **there is no genome-scale system that allows to trace genetic perturbation during scRNA-seq**”.

Cost-saving design choice: make the barcode *be* the guide’s own 20-nt spacer, placed in the 3’UTR of a Pol II reporter. Then no guide-to-barcode linkage sequencing is needed – the mapping is known by construction. This

removes a long-read or paired-end linkage run and a whole class of bookkeeping error, at the price of a slightly more awkward oligo (the spacer appears twice, which a ~200-mer synthesis handles).

What this means for the budget. The library phase is a re-clone, not a re-design: we have the MAGIC designs sha256-pinned already (`experiments/016-lian-magic-reprocess`, 100,493 guides across CRISPRa/i/d; Supplementary Data 1–4 in the mirror). What is needed is a new vector with a Pol II poly(A) reporter, and a fresh oligo pool.

Not priced. Oligo pool synthesis for ~36,000 members at ~100–200 nt is the one material line item with no quote. Lian used CustomArray 92918- and 12472-format chips. **Get a Twist / IDT / GenScript quote before this budget is used for a proposal.** Note also that the pooled MAGIC libraries are **not on Addgene** – only four backbone plasmids (136989, 136992, 136993, 136994) – so there is no shortcut.

Also inherited from MAGIC and worth planning around: guides are constitutive, and Sun notes that in yeast “targeting dosage-sensitive genes such as essential genes resulted in cells with lower fitness once they were transformed”, which is why Momen-Roknabadi built an aTc-inducible version. **Use an inducible system and expand before inducing**, or essential-gene guides drop out before they are measured.

11. Bacteria

Better supported than the yeast case, which is an inversion of the usual situation.

- **microSPLiT is a Nature Protocols method** with 4 barcoding rounds as standard, validated on *B. subtilis* and *E. coli* plus “11 bacterial type species” unpublished.
- **mapSPLiT is the only published pooled single-cell CRISPR screen in any microbe.** It works, in *E. coli* and *P. putida*.
- **The chemistry differs fundamentally, not cosmetically.** Bacterial mRNA is not polyadenylated. microSPLiT solves it with **in-cell poly(A) polymerase**, which “preferentially polyadenylates mRNA as opposed to rRNA” and gives a 2.5x mRNA enrichment. Nothing in a yeast oligo(dT) design carries over.
- **rRNA is worse:** “>95% of all transcripts in *E. coli*”; microSPLiT libraries still end up “~5-10% mRNA”. Brandner’s Cas9-guided depletion took *E. coli* from 4.6% to 60.2% mRNA – but only 2.2% to 10.4% in *P. putida*, so it does not transfer for free between species either.
- Practical floor: microSPLiT wants **>=1 million input cells per sample** and loads 40,000–100,000 cells per RT well, 8–20x the yeast density Brettner used.

Treat bacteria as a **separate development track sharing the barcode plates**, not as a later milestone of the yeast work. The plates, the ligases, and the sequencing recipe are common; the capture chemistry is not.

12. Open questions, ranked by how much they move the budget

1. **What is the yeast mRNA fraction after rRNA depletion?** Everything in §8 turns on it. Brandner’s 13x improvement is an *E. coli* number transferred to yeast. Measure it in the pilot before committing a flow cell. A cheap partial alternative: **shift the polydT:random-hexamer ratio** – Brettner says the random hexamers are what recover rRNA, and for a 3’-counting Perturb-seq the 5’ coverage they buy is not needed.
2. **How many yeast cells fit in an RT well?** Brettner verified 5,000. microSPLiT runs 40,000–100,000 for bacteria. If yeast tolerates even 20,000, the 13 protocol runs in §8 become 4, and the labour argument for 10x largely evaporates.
3. **What is the guide-assignment rate in yeast?** Brandner got 25% in *E. coli*. Yeast has ~4x the mRNA per cell so it should be better, but every point of assignment rate is a point off the sequencing bill.
4. **What does the 4th ligation round cost in molecular yield?** The only argument against it.
5. **Oligo pool synthesis quote** for ~36,000 members (§10).
6. **Does zymolyase spheroplasting distort the transcriptome?** The review is explicit that cell-wall digestion “induc[es] the expression of stress-responsive genes within minutes”, which is why fixation must precede digestion. Brettner fixes first (4% formaldehyde, 18 h). Needs a matched bulk RNA-seq control.

Regenerating everything

```
cd ~/Documents/projects/torchcell    # or the worktree root
python experiments/019-perturb-seq-costing/scripts/cost_model.py
python experiments/019-perturb-seq-costing/scripts/cost_per_cell_table.py
```

```
python experiments/019-perturb-seq-costing/scripts/plot_method_landscape.py
python experiments/019-perturb-seq-costing/scripts/plot_economics.py
```

Outputs land in `experiments/019-perturb-seq-costing/results/` (CSVs) and `$ASSET_IMAGES_DIR/019-perturb-seq-costing/` (SVGs).